Application of WKB Method to Quantum Tunnelling

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Abstract

In this report, the application of the WKB method to quantum tunnelling in one dimension is demonstrated. Probabilities of reflection R and transmission T of a particle with energy E as it encounters a potential barrier are calculated for a delta potential, a general potential $V(x)$ with a single maximum, and for a special case where $E = V_{\max}$ for $V(x) = \exp(-x^2/4)$.

Contents

1 Introduction

Recall from classical physics: if we have a rolling ball with kinetic energy E, and it encounters a potential barrier (consider a hill in this case), it will slow down as it rolls up the hill as some of its kinetic energy is converted to potential energy. If the kinetic energy of the ball is greater than the energy it takes to be at the peak of the potential, then the ball would continue on and roll over the potential barrier. If it did not have enough energy, then it would stop somewhere before the peak and roll back down. This is what is classically possible; the ball can only be where its energy is greater than or equal to the potential barrier.

In quantum mechanics, it is possible for a quantum particle (ie. electron) to "tunnel" through a potential barrier even when its energy is less than that of potential's peak; that is to say, there is a non-zero probability of it appearing on the potential barrier's opposite side even when classically forbidden. This is what we call "quantum tunneling".

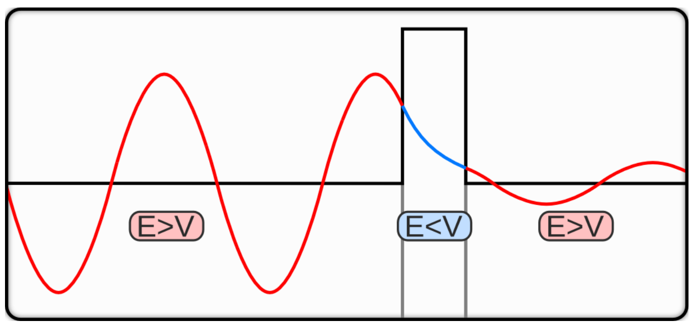

Figure 1: An right-travelling incident wave (originating from left of the barrier) propagating in classically allowed (red) and forbidden regions (blue) as it tunnels through barrier.

To summarize, in a quantum system we have classically allowed regions where $E > V(x)$ and classically forbidden regions where $E < V(x)$; tunneling is when a particle is able to pass through a potential barrier that is classically forbidden. We want to examine this tunneling scenario, so we first study systems for which $E < V_{max}$.

We will proceed by using WKB theory to make a quantitative study of quantum tunneling. Tunneling implicitly involves motion so we will first introduce the notion of waves. Then we examine an exactly solvable model with the Dirac delta potential, and generalize to continuous potentials with a single maximum. We will conclude by looking at a special case where $E = V_{max}$ and scattering occurs.

2 Description of Waves

We begin with the time dependent Schrödinger equation:

$$\frac{1}{i}\frac{\partial}{\partial t}\psi(x,t) = \left[-\epsilon^2\frac{\partial^2}{\partial x^2} + V(x)\right]\psi(x,t) \tag{1}$$

Here $\psi(x,t)$ is the wavefunction of the particle. We can assume its time dependency is purely oscillatory so by separation of variables, it has the following form:

$$\psi(x,t) = y(x)e^{iEt} \tag{2}$$

When we substitute (2) back into (1), we get the following:

$$\frac{1}{i}\frac{\partial}{\partial t}\Big(y(x)e^{iEt}\Big) = \Big[-\epsilon^2\frac{\partial^2}{\partial x^2} + V(x)\Big]\Big(y(x)e^{iEt}\Big) \tag{3}$$

$$\frac{1}{i}y(x)iEe^{iEt} = -\epsilon^2 y''(x)e^{iEt} + V(x)y(x)e^{iEt} \tag{4}$$

$$Ey(x)e^{iEt} = \big[-\epsilon^2 y''(x) + V(x)y(x)\big]e^{iEt} \tag{5}$$

$$Ey(x) = -\epsilon^2 y''(x) + V(x)y(x) \tag{6}$$

$$\epsilon^2 y''(x) = [V(x) - E]y(x) \tag{7}$$

If we let $Q(x) = V(x) - E$, then (7) has the general form of the Schrödinger equation to which we can apply WKB approximations. In the classically allowed regions, $E > V(x)$, we have the following first order WKB approximation:

$$y(x) = C_{\pm}[E - V(x)]^{-1/4}\exp\left[\pm\frac{i}{\epsilon}\int^x\sqrt{E - V(t)}dt\right] \tag{8}$$

Note that in the classically allowed regions, $V(x) - E$ is negative and the square root brings out an i in the exponential, so this solution is oscillatory. To get back the wave function from $y(x)$, we simply multiply it by the time dependent factor:

$$\psi(x,t) = C_{\pm}[E - V(x)]^{-1/4}\exp\left[iEt \pm \frac{i}{\epsilon}\int^x\sqrt{E - V(t)}dt\right] \tag{9}$$

Where the + indicates a left right-moving wave and - indicates a ri-hgtmoving wave as time increases. For the remainder of our analysis, we shall only focus on the spatial component $y(x)$ of the wave function, as it is just a simple multiplication by the time dependent factor to get back the wave function.

3 Delta Functional Potential

3.1 Problem Description

In order to demonstrate the tunnelling phenomenon, we consider an exactly solvable problem where the potential consists of a Dirac delta distribution:

$$\delta(x) = \begin{cases} +\infty & x = 0 \\ 0 & \text{otherwise} \end{cases} \tag{10}$$

Substituting into the Schrödinger equation, we have:

$$\epsilon^2 y''(x) = [V(x) - E]y(x) \tag{11}$$

$$\implies \epsilon^2 y''(x) = [\delta(x) - E]y(x) \tag{12}$$

We assume that a right-travelling incident plane wave arrives from $-\infty$. Tunnelling occurs as the wave approaches $x = 0$ and interacts with the potential barrier. Note that for $x \neq 0$, we have the following:

$$\epsilon^2 y''(x) = Ey(x) \tag{13}$$

This is an exactly solvable ODE. The general solutions are:

$$y(x) = a\exp(-ix\sqrt{E}\epsilon) + b\exp(+ix\sqrt{E}\epsilon) \qquad x < 0 \tag{14}$$

$$y(x) = c\exp(-ix\sqrt{E}\epsilon) + d\exp(+ix\sqrt{E}\epsilon) \qquad x > 0 \tag{15}$$

This corresponds to linear combinations of left- and right-travelling plane waves. For $x > 0$, there is no left-travelling wave as only a right-travelling wave which has tunnelled past the potential barrier can exist. Thus, we set $d = 0$. Additionally, we set $a = 0$ for the incident wave for simplicity. The right-travelling wave in the $x < 0$ region corresponds to the reflected wave, the fraction of the incident wave which has not tunnelled past the potential barrier.

$$y(x) = \exp(-ix\sqrt{E}/\epsilon) + b\exp(+ix\sqrt{E}/\epsilon) \qquad x < 0 \tag{16}$$

$$y(x) = c\exp(-ix\sqrt{E}/\epsilon) \qquad x > 0 \tag{17}$$

3.2 Boundary Conditions

We must match the above equations at $x = 0$. The following conditions must be satisfied:

$$\begin{cases} y(x) \text{ is continuous at } x = 0 \\ \lim\limits_{\eta\to 0} \ \epsilon^2\Big(y'(\eta) - y'(-\eta)\Big) = y(0) \end{cases} \tag{18}$$

The second boundary condition arises due to the potential at $x = 0$. To see why this is true, we introduce the notion of Green's functions. By definition, the Green's function G corresponding to a linear differential operator L has the property that $LG(x) = \delta(x)$; this is precisely the property required of the solution at $x = 0$.

For a general Schrödinger equation, consider the following Green's function equation for some fixed x':

$$\epsilon^2\frac{\partial^2}{\partial x^2}G(x,x') - Q(x)G(x-x') = -\delta(x-x') \tag{19}$$

Applying the WKB approximation for region I ($x' < x$) and region II ($x' > x$):

$$G_{\mathrm{I}}(x,x') = \mathrm{C_I}\exp\left[-\frac{1}{\epsilon}\int_{x'}^{x}\sqrt{Q(t)}dt\right] \quad \text{as x} \quad \to +\infty \tag{20}$$

$$G_{\mathrm{II}}(x,x') = \mathrm{C_{II}}\exp\left[-\frac{1}{\epsilon}\int_{x}^{x'}\sqrt{Q(t)}dt\right] \quad \text{as x} \quad \to -\infty \tag{21}$$

We require the solution to be continuous at $x = 0$. Hence:

$$\lim_{\eta\to 0^+}\Big[G_{\mathrm{I}}(x+\eta,x') - G_{\mathrm{II}}(x'-\eta,x')\Big] = 0 \tag{22}$$

At $x = x'$, we integrate from $x'+\eta$ and $x'-\eta$ and take the limit that $\eta \to 0^+$ to obtain the following:

$$\lim_{\eta\to 0^+}\left[\frac{\partial}{\partial x}G_{\mathrm{I}}(x,x')|_{x=x'+\eta} - \frac{\partial}{\partial x}G_{\mathrm{II}}(x,x')|_{x=x'-\eta}\right] = -\frac{1}{\epsilon^2} \tag{23}$$

This gives us boundary condition (18) as desired.

3.3 Transmission and Reflection Coefficients

The transmission and reflection coefficients can be obtained by applying the boundary conditions in equations (18) together with (16) and (17). This allows us to write the following:

$$c = 1 + b \tag{24}$$

$$\lim_{\eta\to 0^+} \epsilon^2 \left[-i\frac{\sqrt{E}}{\epsilon} c \exp\left(-i\eta\frac{\sqrt{E}}{\epsilon}\right) + i\frac{\sqrt{E}}{\epsilon} c \exp\left(i\eta\frac{\sqrt{E}}{\epsilon}\right) - i\frac{\sqrt{E}}{\epsilon} b \exp\left(-i\eta\frac{\sqrt{E}}{\epsilon}\right)\right] = 0 \tag{25}$$

Thus we are able to conclude by solving the above system that:

$$b = \frac{2\epsilon\sqrt{E}i - 1}{4\epsilon^2 E + 1} \tag{26}$$

$$c = \frac{2\epsilon\sqrt{E}(2\epsilon\sqrt{E} + i)}{4\epsilon^2 E + 1} \tag{27}$$

Comparing this to (16) and (17), we see that b corresponds to the probability of an incident particle with energy E being reflected, whereas c corresponds to the probability of the incident particle with energy E being transmitted, i.e. "tunnels" past the potential barrier. The total probability should add up to $R + T = 1$. The reflection coefficient R and transmission coefficient T are defined as follows:

$$R = |b|^2 = \frac{1}{4\epsilon^2 E + 1} \tag{28}$$

$$T = |c|^2 = \frac{4\epsilon^2 E}{4\epsilon^2 E + 1} \tag{29}$$

Thus there is a non-zero chance that an incident particle will be found right of the potential barrier. We also make the following observations:

$$\begin{cases} E = (4\epsilon^2)^{-1} & \implies \text{R} = 1/2, \text{T} = 1/2 \\ E < (4\epsilon^2)^{-1} & \implies \text{R} > \text{T} \\ E > (4\epsilon^2)^{-1} & \implies \text{R} < \text{T} \end{cases} \tag{30}$$

In the quantum case, both T and R can be non-negligible. However, classically the energy E of particles is either defined to be 0 (ie. if it is at rest) or large in relation to quantum particles (ie. if it is moving at a macroscopically appreciable speed). This is especially true if ϵ is very small, as is the case in the tunnelling problem. Thus we usually have that:

$$\begin{cases} E_{\text{classical}} \ll (4\epsilon^2)^{-1} & \implies \text{R} = 1, \text{T} = 0 \\ E_{\text{classical}} \gg (4\epsilon^2)^{-1} & \implies \text{R} = 0, \text{T} = 1 \end{cases} \tag{31}$$

Contrary to the probabilistic tunnelling behavior of quantum particles, classical particles either overcome the potential barrier or they do not. This confirms our prior expectations.

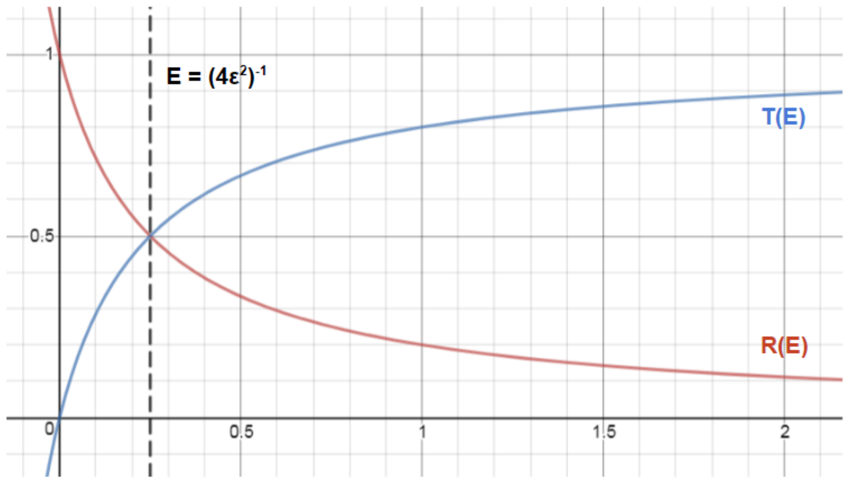

Figure 2: Plot of T, R dependence on E with $\epsilon = 1$ for demonstrative purposes.

4 WKB Description of Tunnelling

4.1 Problem Description

We now generalize from our previous delta potential model. Let $V(x)$ be any continuous function that vanishes at $x = \pm\infty$ and increases monotonically to its maximum at $x = x'$ from either side. Let there be two turning points A and B, where $A < B$ and $V(A) = V(B) = E$. There are now two classically allowed regions ($x < A$ and $x > B$) where oscillatory solutions exist, and a classically forbidden region ($A < x < B$) where an exponentially decaying solution exists. We also assume as $x \to \pm\infty$, $V(x) \to 0$ faster than $\frac{1}{x}$. We then can make the following approximation on the eikonal equation $S_o(x) = \pm \int^x \sqrt{Q(t)}dt$ in the classically allowed region $x > B$:

$$\int_B^x dt\sqrt{E - V(t)} = \int_B^x dt\Big[\sqrt{E - V(t)} - \sqrt{E} + \sqrt{E}\Big] \tag{32}$$

$$= \sqrt{E}\int_B^x dt + \int_B^x dt\Big[\sqrt{E - V(t)} - \sqrt{E}\Big] \tag{33}$$

$$= x\sqrt{E} - B\sqrt{E} + \int_B^x dt\Big[\sqrt{E - V(t)} - \sqrt{E}\Big] \tag{34}$$

$$\sim x\sqrt{E} + I, \quad \text{as} \quad x \to +\infty \tag{35}$$

$$\text{where} \quad I = \int_B^\infty dt\Big[\sqrt{E - V(t)} - \sqrt{E}\Big] - B\sqrt{E} \tag{36}$$

And similarly for $x < A$:

$$\int_x^A dt\sqrt{E-V(t)} = \int_x^A dt\Big[\sqrt{E-V(t)} - \sqrt{E} + \sqrt{E}\Big] \tag{37}$$

$$= \sqrt{E}\int_x^A dt + \int_x^A dt\Big[\sqrt{E-V(t)} - \sqrt{E}\Big] \tag{38}$$

$$= A\sqrt{E} - x\sqrt{E} + \int_x^A dt\Big[\sqrt{E-V(t)} - \sqrt{E}\Big] \tag{39}$$

$$\sim -x\sqrt{E} + J, \quad \text{as} \quad x \to -\infty \tag{40}$$

$$\text{where} \quad J = \int_{-\infty}^A dt\Big[\sqrt{E-V(t)} - \sqrt{E}\Big] + A\sqrt{E} \tag{41}$$

With this, we can make the following approximations on the oscillatory WKB solutions in the classically allowed regions:

For $x > B$:

$$y(x) = C_\pm[E-V(x)]^{-1/4}\exp\left[\pm\frac{i}{\epsilon}\int_B^x \sqrt{E-V(t)}dt\right] \tag{42}$$

$$\sim C_\pm E^{-1/4}e^{\pm iI/\epsilon}\exp\Big[\pm ix\sqrt{E}/\epsilon\Big], \quad \text{as} \quad x \to +\infty \tag{43}$$

For $x < A$:

$$y(x) = D_\pm[E-V(x)]^{-1/4}\exp\left[\pm\frac{i}{\epsilon}\int_x^A \sqrt{E-V(t)}dt\right] \tag{44}$$

$$\sim D_\pm E^{-1/4}e^{\pm iJ/\epsilon}\exp\Big[\mp ix\sqrt{E}/\epsilon\Big], \quad \text{as} \quad x \to -\infty \tag{45}$$

Therefore, wee see that the oscillatory WKB solutions in these classically allowed regions can be approximated as plane waves as $x \to \pm\infty$, like in the delta potential model.

4.2 Boundary Conditions

Again, we choose the appropriate boundary conditions to describe tunneling. Suppose we have a unit incident right-moving wave at $x = -\infty$ traveling towards $x = x'$. This should give rise to a right-moving transmitted wave at $x = +\infty$, and a reflected left-moving wave at $x = -\infty$. There is no left traveling wave on the right side of $x = x'$. So the asymptotic plane wave approximations should look the same as before:

$$y(x) = \exp(-ix\sqrt{E}/\epsilon) + b\exp(+ix\sqrt{E}/\epsilon) \qquad x \to -\infty \tag{46}$$

$$y(x) = c\exp(-ix\sqrt{E}/\epsilon) \qquad x \to +\infty \tag{47}$$

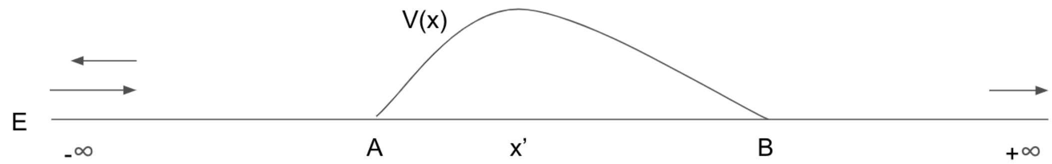

Figure 3: Two turning point problem with boundary conditions.

Note that for the approximations (46) and (47) to agree with boundary conditions (43) and (45), we have:

$$D_+ E^{-1/4} e^{+iJ/\epsilon} = 1 \quad \Rightarrow D_+ = E^{1/4} e^{-iJ/\epsilon} \tag{48}$$
$$D_- E^{-1/4} e^{-iJ/\epsilon} = b \quad \Rightarrow D_- = bE^{1/4} e^{iJ/\epsilon} \tag{49}$$
$$C_- E^{-1/4} e^{-iI/\epsilon} = c \quad \Rightarrow C_- = cE^{1/4} e^{iI/\epsilon} \tag{50}$$
$$C_+ E^{-1/4} e^{+iI/\epsilon} = 0 \quad \Rightarrow C_+ = 0 \tag{51}$$

4.3 Connection Formula

We begin our analysis for this problem by decomposing the two-turning point problem into two one-turning point problems. We focus (ie. "zoom in") on one of the two turning points, $x = B$, and shift it to the origin, so that for $\epsilon^2 y''(x) = Q(x)y(x)$ we have the following conditions on $Q(x)$:

$$\epsilon^2 y''(x) = Q(x)y(x) \qquad \begin{cases} Q(0) = 0 \\ Q(x) < 0 & x > 0 \\ Q(x) > 0 & x < 0 \\ Q(x) \sim ax & x \to 0, \quad a < 0 \end{cases} \tag{52}$$

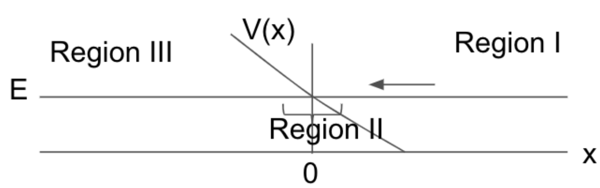

Figure 4: One turning point problem.

In region I, $x > 0$, if the general oscillatory WKB solution has negative phase:

$$y_I(x) = [-Q(x)]^{-1/4} \exp\left[-\frac{i}{\epsilon} \int_0^x \sqrt{-Q(t)}dt \right] \tag{53}$$

Then we want to know how $y(x)$ changes as x becomes negative. First, we take $x \to 0^+$ in equation (53) and we get:

$$y_I(x) \sim [-ax]^{-1/4} \exp\left[-\frac{i}{\epsilon} \int_0^x \sqrt{-at}dt \right] \tag{54}$$

$$= [-ax]^{-1/4} \exp\left[-\frac{i\sqrt{-a}}{\epsilon} \int_0^x t^{1/2}dt \right] \tag{55}$$

$$= [-ax]^{-1/4} \exp\left(-\frac{2i\sqrt{-a}}{3\epsilon} x^{3/2} \right) \qquad x \to 0^+ \tag{56}$$

Then we look at region II where $Q(x) \sim ax$. We make the following change of variable to (52):

$$t = \left(\frac{-a}{\epsilon^2} \right)^{1/3} x \tag{57}$$

And we get the Airy equation $y'' = -ty$, which has the general solution:

$$y_{\text{II}}(t) = \alpha\ Ai(-t) + \beta\ Bi(-t) \tag{58}$$

where Ai is the Airy function of the first kind and Bi is the Airy function of the second kind.

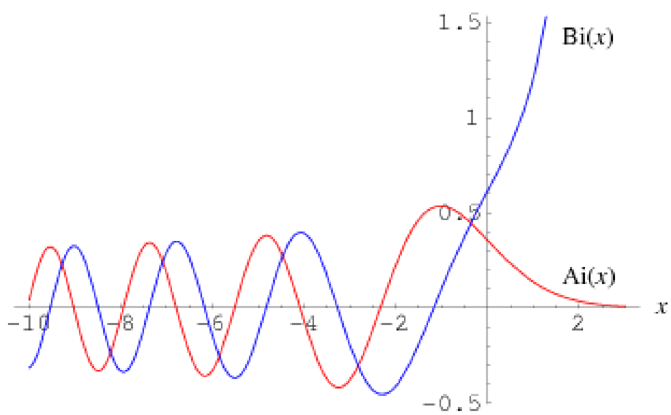

Figure 5: Plot of the Airy functions.

When $t \to +\infty$, i.e. when $\epsilon \to 0^+$ and $x > 0$, we can replace Ai and Bi in (58) with the following approximations:

$$Ai(-t) \sim \frac{1}{\sqrt{\pi}} t^{-1/4} \sin\left(\frac{2}{3} t^{3/2} + \frac{\pi}{4} \right), \qquad t \to +\infty \tag{59}$$

$$Bi(-t) \sim \frac{1}{\sqrt{\pi}} t^{-1/4} \cos\left(\frac{2}{3} t^{3/2} + \frac{\pi}{4} \right), \qquad t \to +\infty \tag{60}$$

$$(58) \Rightarrow y_{\text{II}}(t) \sim \frac{\alpha}{\sqrt{\pi}} t^{-1/4} \sin\left(\frac{2}{3} t^{3/2} + \frac{\pi}{4} \right) + \frac{\beta}{\sqrt{\pi}} t^{-1/4} \cos\left(\frac{2}{3} t^{3/2} + \frac{\pi}{4} \right), \quad t \to +\infty \tag{61}$$

If we plug (57) back into the above equation and compare the with equation (56), we find the coefficients to be:

$$\alpha = \sqrt{\pi}(-\epsilon a)^{-1/6} e^{-i\pi/4} \tag{62}$$

$$\beta = \sqrt{\pi}(-\epsilon a)^{-1/6} e^{i\pi/4} \tag{63}$$

Now we let $t \to -\infty$, ($\epsilon \to 0^+$, $x < 0$), and find that Ai(-t) is negligible compared to Bi(-t), as can be seen in Figure (5). We use the following expansion for Bi:

$$Bi(-t) \sim \frac{1}{\pi}(-t)^{-1/4} e^{\frac{2}{3}(-t)^{3/2}}, \quad t \to -\infty \tag{64}$$

and the coefficients α and β to find that for region III:

$$y_{\rm III}(x) = [Q(x)]^{-1/4} \exp\left[\frac{i\pi}{4} + \frac{1}{\epsilon}\int_x^0 \sqrt{Q(t)}dt\right] \tag{65}$$

We can summarize our results with a connection formula that tells us how the solution changes from region I to region III (the arrow indicates the direction of the connection):

$$[Q(x)]^{-1/4} \exp\left[\frac{i\pi}{4} + \frac{1}{\epsilon}\int_x^0 \sqrt{Q(t)}dt\right], \quad x < 0 \tag{66}$$

$$\leftarrow [-Q(x)]^{-1/4} \exp\left[-\frac{i}{\epsilon}\int_0^x \sqrt{-Q(t)}dt\right], \quad x > 0 \tag{67}$$

If we zoom in on the turning point at $x = A$ and shift it to the origin, then we get a one turning point problem we've seen before where $Q(0) = 0$, $Q(x) > 0$ for $x > 0$, $Q(x) < 0$ for $x < 0$, and $Q(x) \sim ax$ for $a > 0$ as $x \to 0$. We can follow a similar derivation as above and get the connection formula [1] (10.4.16):

$$2[-Q(x)]^{-1/4} \sin\left[\frac{\pi}{4} + \frac{1}{\epsilon}\int_x^0 \sqrt{-Q(t)}dt\right], \quad x < 0 \tag{68}$$

$$\leftarrow [Q(x)]^{-1/4} \exp\left[-\frac{1}{\epsilon}\int_0^x \sqrt{Q(t)}dt\right], \quad x > 0 \tag{69}$$

It will be useful to recall Euler's identity for sine function: $\sin\theta = \frac{e^{i\theta} - e^{-i\theta}}{2i}$.

4.4 Two Turning Point Problem

Now we return to our two turning point problem.

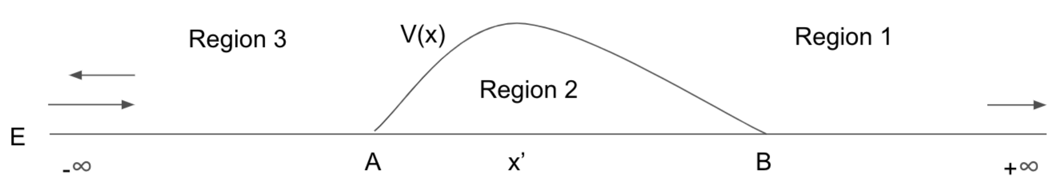

Figure 6: Regions of the two turning point problem.

In region 1, $x > B$, we have from equation (42) and (50) that the WKB solution is:

$$y_1(x) = ce^{iI/\epsilon}\left[\frac{E - V(x)}{E}\right]^{-1/4} \exp\left[-\frac{i}{\epsilon}\int_B^x \sqrt{E - V(t)}dt\right] \tag{70}$$

By the connection formula (67) to (66), the solution in region 2, $A < x < B$, connecting from (70) is:

$$y_2(x) = ce^{iI/\epsilon}\left[\frac{V(x)-E}{E}\right]^{-1/4} \exp\left[\frac{i\pi}{4} + \frac{1}{\epsilon}\int_x^B \sqrt{V(t)-E}dt\right] \tag{71}$$

Which we can rearrange into:

$$y_2(x) = ce^{iI/\epsilon}\left[\frac{V(x)-E}{E}\right]^{-1/4} \exp\left[\frac{i\pi}{4} + \frac{1}{\epsilon}\left(\int_A^B \sqrt{V(t)-E}dt - \int_A^x \sqrt{V(t)-E}dt\right)\right] \tag{72}$$

$$= ce^{iI/\epsilon}\exp\left[\frac{i\pi}{4} + \frac{1}{\epsilon}\int_A^B \sqrt{V(t)-E}dt\right]\left[\frac{V(x)-E}{E}\right]^{-1/4} \exp\left[-\frac{1}{\epsilon}\int_A^x \sqrt{V(t)-E}dt\right] \tag{73}$$

Note that this solution decays exponentially as $x \to B$. Now we use the other connection formula (69) to (68) to get the solution in the final region 3, $x < A$, from equation (73):

$$\begin{aligned} y_3(x) = ce^{iI/\epsilon}\exp&\left[\frac{1}{\epsilon}\int_A^B \sqrt{V(t)-E}dt\right]\left[\frac{E-V(x)}{E}\right]^{-1/4} \\ &\times\left\{\exp\left[\frac{i}{\epsilon}\int_x^A \sqrt{E-V(t)}dt\right] + i\exp\left[-\frac{i}{\epsilon}\int_x^A \sqrt{E-V(t)}dt\right]\right\} \end{aligned} \tag{74}$$

We can take $x \to -\infty$, where $V(x) \to 0$, and we have the following approximation:

$$\begin{aligned} y_3(x) \sim ce^{iI/\epsilon}\exp&\left[\frac{1}{\epsilon}\int_A^B \sqrt{V(t)-E}dt\right] \\ &\times\left[e^{iJ/\epsilon}\exp\left(-i\sqrt{E}x/\epsilon\right) + ie^{-iJ/\epsilon}\exp\left(i\sqrt{E}x/\epsilon\right)\right], \quad x \to -\infty \end{aligned} \tag{75}$$

Which is similar to the plane wave approximation in equation (45).

4.5 Transmission and Reflection Coefficients

When we compare the WKB solution (75) found in the previous section with equations (45), (46) and the conditions (48) and (49), we obtain that:

$$1 = ce^{iI/\epsilon}\exp\left[\frac{1}{\epsilon}\int_A^B \sqrt{V(t)-E}dt\right]e^{iJ/\epsilon} \tag{76}$$

$$\Rightarrow c = \exp\left[-\frac{1}{\epsilon}\int_A^B \sqrt{V(t)-E}dt\right]e^{-iI/\epsilon}e^{-iJ/\epsilon} \tag{77}$$

$$= \exp\left[-\frac{1}{\epsilon}\int_A^B \sqrt{V(t)-E}dt\right]e^{-i(I+J)/\epsilon} \tag{78}$$

And:

$$
\begin{aligned}
b &= ce^{iI/\epsilon} \exp\left[\frac{1}{\epsilon}\int_A^B \sqrt{V(t)-E}dt\right] ie^{-iJ/\epsilon} && (79)\\
&= \exp\left[-\frac{1}{\epsilon}\int_A^B \sqrt{V(t)-E}dt\right] e^{-iI/\epsilon}e^{-iJ/\epsilon}e^{iI/\epsilon} \exp\left[\frac{1}{\epsilon}\int_A^B \sqrt{V(t)-E}dt\right] ie^{-iJ/\epsilon} && (80)\\
&= ie^{-2iJ/\epsilon} && (81)
\end{aligned}
$$

Therefore the reflection coefficient R is:

$$R = |b^2| \sim 1, \quad \epsilon \to 0^+ \tag{82}$$

And the transmission coefficient T is:

$$T = |c^2| \sim \exp\left[-\frac{2}{\epsilon}\int_A^B \sqrt{V(t)-E}dt\right], \quad \epsilon \to 0^+ \tag{83}$$

This demonstrates there is a small nonzero probability that the wave will tunnel through the potential barrier. The leading behavior of R is approximately 1 but not exactly 1; similarly, the leading behavior of T is small but non-negligible compared to zero. This ensures that $R + T = 1$, by definition.

5 Scattering off the Potential Peak

5.1 Problem Description

In the previous analysis, the result is general but valid only when $E < V_{\max}(x)$. The situation where $E = V_{\max}(x)$ corresponds to scattering off of the potential peak and must be studied separately. Note that substituting $E = V_{\max}(x)$ into the result from the prior section we obtain $T = 1$ and $R = 1$, which is unphysical.

Consider the following problem, which we will use as a representative example and solve in full:

$$
\begin{cases}
V(x) = \exp(-x^2/4)\\
V_{\max} = 1\\
E = 1
\end{cases} \tag{84}
$$

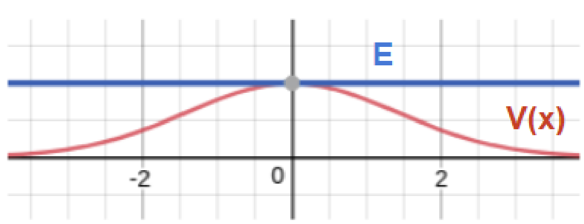

Figure 7: Setup of scattering off potential peak problem.

Substituting into the Schrödinger equation, we have that:

$$[V(x) - E]y(x) = \epsilon^2 y''(x) \tag{85}$$
$$\implies \Big(- \epsilon^2 \frac{d^2}{dx^2} + \exp\frac{-x^2}{4} - 1 \Big) y(x) = 0 \tag{86}$$

To solve this problem, we divide the the real line $-\infty < x < +\infty$ into three regions, solving the appropriate Schrödinger in each and matching at the boundaries:

$$\begin{cases} \text{Region } \ \text{I} : x > 0 \\ \text{Region } \ \text{II} : x \sim 0 \\ \text{Region } \ \text{III} : x < 0 \end{cases} \tag{87}$$

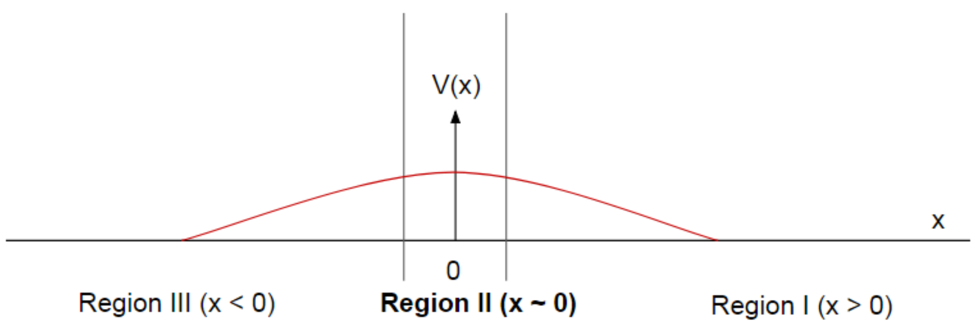

Figure 8: Regions of the scattering off peak problem.

5.2 Region II ($x \sim 0$)

We begin the analysis by studying the region where x $\sim$ 0; this will be used to connect the solutions in region I and III. Call the solution in this region $y_{\text{II}}(x)$. In region II $|x|$ is small, so it is appropriate to take the Taylor expansion of $V(x) = \exp(-x^2/4) \sim 1 - x^2/4$. Substituting into equation (86), we obtain:

$$\Big(- \epsilon^2 \frac{d^2}{dx^2} - \frac{x^2}{4} \Big) y_{\text{II}}(x) = 0 \tag{88}$$

We notice that this is of the form of a parabolic cylinder equation (also known as *Weber's equation*), the solution of which is a linear combination of parabolic cylinder functions D_v. In the case where $v \neq 0, 1, 2...$ we have that:

$$\Big(- a \frac{d^2}{dt^2} - \frac{t^2}{4} - v - \frac{1}{2} \Big) z(t) = 0 \tag{89}$$
$$\implies z(t) = \alpha D_v \Big(\frac{-1^{\frac{1}{4}} x}{a^{\frac{1}{4}}} \Big) + \beta D_v \Big(\frac{-1^{\frac{3}{4}} x}{a^{\frac{1}{4}}} \Big) \tag{90}$$

Comparing equation (88) to (89), we see that $v = -1/2$ to eliminate the $-1/2$ term. Thus we are able to conclude that:

$$y_{\mathrm{II}}(x) = \alpha D_{1/2}\Big(e^{-i\pi/4}\frac{x}{\sqrt{\epsilon}}\Big) + \beta D_{-1/2}\Big(-e^{-i\pi/4}\frac{x}{\sqrt{\epsilon}}\Big) \tag{91}$$

In the above we have rewritten the roots of unity in complex exponential notation using Euler's formula. To investigate the behavior of $y_{\mathrm{II}}(x)$ at the left and right boundary of region II, we consider the asymptotic behavior of the parabolic cylinder functions $D_v(t)$ for $t = x/\sqrt{\epsilon} \to \pm\infty$ (this is valid as we have taken $V(x)$ in this region to be the Taylor expansion for small x, so taking the limit of $t \to \pm\infty$ will still produce the relevant solutions for region II). To first order, we have that:

$$D_v(t) = t^v e^{-t^2/4} \qquad t \to \infty, \quad |\arg(t)| < \frac{3\pi}{4} \tag{92}$$

$$D_v(t) = t^v e^{-t^2/4} - \frac{\sqrt{2\pi}}{\Gamma(-v)} e^{-i\pi v} t^{-v-1} e^{t^2/4} \qquad t \to \infty, \quad \frac{\pi}{4} < |\arg(t)| < \frac{5\pi}{4} \tag{93}$$

Referring to (91), we note that $\arg(e^{-i\pi/4}x/\sqrt{\epsilon}) = i\pi/4$ and $\arg(-e^{-i\pi/4}x/\sqrt{\epsilon}) = 3i\pi/4$; this determines which of the above two asymptotic approximations to apply for each term. Substituting $v = -1/2$, $t = x/\sqrt{\epsilon}$ and recalling that $\Gamma(-1/2) = \sqrt{\pi}$, we obtain:

$$y_{\mathrm{II}} \sim \epsilon^{1/4} x^{-1/2}\big[(\alpha e^{i\pi/8} + \beta e^{-3i\pi/8})e^{ix^2/4\epsilon} + \beta\sqrt{2}e^{i\pi/8}e^{-ix^2/4\epsilon}\big] \qquad x/\sqrt{\epsilon} \to +\infty \tag{94}$$

$$y_{\mathrm{II}} \sim \epsilon^{1/4} (-x)^{-1/2}\big[(\alpha e^{-i\pi/8} + \beta e^{3i\pi/8})e^{ix^2/4\epsilon} + \alpha\sqrt{2}e^{i\pi/8}e^{-ix^2/4\epsilon}\big] \qquad x/\sqrt{\epsilon} \to -\infty \tag{95}$$

The solutions in region I must match to (94) as $x \to 0+$ and the solutions in region III must match to (115) as $x \to 0-$.

5.3 Region I ($x > 0$)

Consider now the region $x > 0$. Here we apply the WKB approximation with $Q(x) = E - V(x) = 1 - \exp(-x^2/4)$, so that we obtain:

$$y_{\mathrm{I}} = A(1 - e^{-x^2/4})^{-1/4} \exp\Big(-\frac{i}{\epsilon}\int_0^x dt\sqrt{1 - e^{-t^2/4}}\Big) \tag{96}$$

The negative exponent solution corresponds to a right-travelling wave which has tunnelled past the potential barrier. Similar to the analysis for (17), no left-travelling wave can exist in this region.

We are interested in the asymptotic behavior of $y_{\mathrm{I}}(x)$ as $x \to \pm\infty$, which will allow us to match boundary conditions. We first study the case of $x \to +\infty$, where we have:

$$(1 - e^{-x^2/4})^{-1/4} \sim 1 \qquad x \to +\infty \tag{97}$$

$$-\frac{i}{\epsilon}\int_0^x dt\sqrt{1 - e^{-t^2/4}} \sim -\frac{i}{\epsilon}\Big(\int_0^\infty dt\sqrt{1 - e^{-t^2/4}} + \int_0^x dt\sqrt{1 - e^{-t^2/4}}\Big) \qquad x \to +\infty \tag{98}$$

$$\sim -\frac{i}{\epsilon}\Big(\int_0^\infty dt\sqrt{1 - e^{-t^2/4}} + \int_0^x dt(1)\Big) \tag{99}$$

$$= -\frac{i}{\epsilon}\Big(\int_0^\infty dt\sqrt{1 - e^{-t^2/4}} + x\Big) \tag{100}$$

For simplicity, let $I = \int_0^\infty dt\sqrt{1 - e^{-t^2/4}}$. Thus we have that: $y_{\mathrm{I}}(x) \sim Ae^{-iI/\epsilon}e^{-ix/\epsilon}$ Recalling 17 from the delta function example, we expect the solution to approach the form of plane waves far from

$x = 0$ (since $V(x)$ is exponentially decaying and its effect becomes negligible); this is the situation observed here. Thus:

$$c = Ae^{-iI/\epsilon} \tag{101}$$

Now consider the limit of $x \to 0+$. We have that $V(x) = \exp(-x^2/4) \sim 1 - x^2/4$. Thus, the terms in $y_{\mathrm{I}}(x)$ become:

$$\begin{aligned}
(1 - e^{-x^2/4})^{-1/4} &\sim (1 - (1 - x^2/4))^{-1/4} && x \to 0+ && (102)\\
&= (2/x)^{1/2} && && (103)\\
-\frac{i}{\epsilon}\int_0^x dt\sqrt{1 - e^{-t^2/4}} &\sim -\frac{i}{\epsilon}\Big(\int_0^x dt\sqrt{1 - (1 - x^2/4)}\Big) && x \to +\infty && (104)\\
&= -\frac{i}{\epsilon}\Big(\int_0^x dt\Big(\frac{x}{2}\Big)\Big) && && (105)\\
&= -\frac{ix^2}{4\epsilon} && && (106)
\end{aligned}$$

Combining the above results we have that $y_{\mathrm{I}}(x) \sim A(2/x)^{1/2}e^{-ix^2/4\epsilon}$. This must match with the solution obtained earlier for the right (ie. positive x) limit of region II, which we restate here:

$$y_{\mathrm{II}} \sim \epsilon^{1/4}x^{-1/2}\big[(\alpha e^{i\pi/8} + \beta e^{-3i\pi/8})e^{ix^2/4\epsilon} + \beta\sqrt{2}e^{i\pi/8}e^{-ix^2/4\epsilon}\big] \qquad x/\sqrt{\epsilon} \to +\infty \tag{107}$$

Comparing the two expressions, we see that $y_{\mathrm{I}}(x)$ contains a $e^{-ix^2/4\epsilon}$ term whereas $y_{\mathrm{II}}(x)$ contains a $e^{-ix^2/4\epsilon}$ and $e^{ix^2/4\epsilon}$ term. To match it must be that:

$$\epsilon^{1/4}\beta e^{i\pi/8} = A \tag{108}$$

$$\alpha e^{i\pi/8} + \beta e^{-3i\pi/8} = 0 \tag{109}$$

This concludes our analysis for region I. Complete determination of the boundary conditions requires analysis of region III.

5.4 Region III ($x < 0$)

The analysis for this section is similar to that for region I. Using the WKB approximation for $x < 0$, we obtain that:

$$y_{\mathrm{I}} = (1 - e^{-x^2/4})^{-1/4}\left[B\exp\Big(\frac{i}{\epsilon}\int_x^0 dt\sqrt{1 - e^{-x^2/4}}\Big) + C\exp\Big(\frac{-i}{\epsilon}\int_x^0 dt\sqrt{1 - e^{-x^2/4}}\Big)\right] \tag{110}$$

Here we keep solutions of both signs: the negative solution corresponds to the right-travelling incident wave, whereas the positive solution corresponds to the reflected wave.

Consider the limit $x \to -\infty$; here the contribution of $V(x)$ is negligible, so we expect the solution to be left- and right-travelling plane waves as (16). Using the same approximation as in (100), we obtain that:

$$y_{\mathrm{III}} \sim Be^{-iI/\epsilon}e^{-ix/\epsilon} + Ce^{-iI/\epsilon}e^{ix/\epsilon} \tag{111}$$

Where $I = \int_0^\infty dt\sqrt{1-e^{-t^2/4}}$ as before. Comparing this to (16), we have that:

$$Be^{-iI/\epsilon} = 1 \tag{112}$$
$$Ce^{-iI/\epsilon} = b \tag{113}$$

Now consider the limit $x \to 0-$. Again, $V(x) = \exp(-x^2/4) \sim 1 - x^2/4$ as x is small in this limit. Using the same approximations as (106), we obtain that:

$$y_{\mathrm{III}} \sim (-2/x)^{1/2}(Be^{ix^2/4\epsilon} + Ce^{-ix^2/4\epsilon}) \tag{114}$$

This solution must match with the result for the left (ie. negative x) limit of region II, which we restate here:

$$y_{\mathrm{II}} \sim \epsilon^{1/4}(-x)^{-1/2}\big[(\alpha e^{-i\pi/8} + \beta e^{3i\pi/8})e^{ix^2/4\epsilon} + \alpha\sqrt{2}e^{i\pi/8}e^{-ix^2/4\epsilon}\big] \qquad x/\sqrt{\epsilon} \to -\infty \tag{115}$$

Comparing the two, we can see by inspection that the following relations must be true:

$$\epsilon^{1/4}\alpha e^{i\pi/8} = C \tag{116}$$
$$\epsilon^{-1/4}(\alpha e^{-3i\pi/8} + \beta e^{i\pi/8}) = B\sqrt{2} \tag{117}$$

This is the last set of boundary conditions which will be necessary to obtain the reflection and transmission coefficients.

5.5 Transmission and Reflection Coefficients

The coefficients b and c related to R and T are given by applying equations (101), (108), (109), (112), (113), (116), (117). We repeat them again here:

$$\begin{cases} Ae^{-iI/\epsilon} & = c \qquad (101) \\ \epsilon^{1/4}\beta e^{i\pi/8} & = A \qquad (108) \\ \alpha e^{i\pi/8} + \beta e^{-3i\pi/8} & = 0 \qquad (109) \\ Be^{-iI/\epsilon} & = 1 \qquad (112) \\ Ce^{-iI/\epsilon} & = b \qquad (113) \\ \epsilon^{1/4}\alpha e^{i\pi/8} & = C \qquad (116) \\ \epsilon^{-1/4}(\alpha e^{-3i\pi/8} + \beta e^{i\pi/8}) & = B\sqrt{2} \qquad (117) \end{cases} \tag{118}$$

From the above we can readily determine through algebra that $c = \frac{1}{\sqrt{2}}$ and $b = \frac{i}{\sqrt{2}}$. Thus the values for the coefficients b and c in full are:

$$b = \frac{i}{\sqrt{2}}\exp\Big(-\frac{2i}{\epsilon}\int_0^\infty dt\sqrt{1-e^{-t^2/4}} - 1\Big) \tag{119}$$
$$c = \frac{1}{\sqrt{2}}\exp\Big(-\frac{2i}{\epsilon}\int_0^\infty dt\sqrt{1-e^{-t^2/4}} - 1\Big) \tag{120}$$

From this we obtain that:

$$R = |b|^2 = \frac{1}{2} \tag{121}$$

$$T = |c|^2 = \frac{1}{2} \tag{122}$$

This agrees with intuition; for a symmetric $V(x)$, there is an equal probability of reflection and transmission when scattering off of the potential peak. For different a $V(x)$ the calculation must be performed again from the beginning.

6 Conclusion

The quantum tunnelling effect is studied using the WKB method on the Schrödinger equation:

$$\epsilon^2 y''(x) = [V(x) - E]y(x) \tag{123}$$

An exactly solvable problem with $V(x) = \delta(x)$ was considered and the following results for transmission and reflection were found:

$$R = |b|^2 = \frac{1}{4\epsilon^2 E + 1} \tag{124}$$

$$T = |c|^2 = \frac{4\epsilon^2 E}{4\epsilon^2 E + 1} \tag{125}$$

We then generalized to any continuous potential function $V(x)$ which vanished at $x = \pm\infty$ and rose monotonically to its max from either side. This became a two turning point problem, which we solved using the WKB method and found the coefficients of reflection and transmission to be:

$$R \sim 1, \quad \epsilon \to 0^+ \tag{126}$$

$$T \sim \exp\left[-\frac{2}{\epsilon} \int_A^B \sqrt{V(t) - E}dt \right], \quad \epsilon \to 0^+ \tag{127}$$

In the case of $E = V_{\max}$, we studied the scattering effect of an incident particle wave off of the potential peak for $V(x) = \exp(-x^2/4)$ and $E = 1$. For this symmetric potential, the transmission and reflection probabilities were shown to be equal at half and half, which agrees with physical intuition.

References

[1] Carl M. Bender, Steven A. Orszag. *Advanced Mathematical Methods for Scientists and Engineers I: Asymptotic Methods and Perturbation Theory.* Springer, 1999

[2] David J. Griffiths. *Introduction to Quantum Mechanics.* 2^{nd} Edition, Pearson, 2005